Available online at www.sciencedirect.com

SCIENCE DIRECT®

American Journal of OTOLARYNGOLOGY

American Journal of Otolaryngology–Head and Neck Medicine and Surgery 26 (2005) 302–307

www.elsevier.com/locate/amjoto

Biofeedback training of nasal muscles using internal and external surface electromyography of the nose

Michael Vaiman, MD, PhD*, Nathan Shlamkovich, MD, Alex Kessler, MD, Ephraim Eviatar, MD, Samuel Segal, MD

Department of Otolaryngology, Assaf Harofeh Medical Center[1], Tel Aviv University, Tel Aviv, Israel

Revised 9 August 2004

Abstract

Background: The purpose of this paper is to describe the outcome of biofeedback training of nasal muscles in cases of nasal valve stenosis and collapse. The present study was performed to investigate the best way of using surface electromyography (sEMG) in biofeedback training of muscles involved in nasal valve function. In the present study, we present the way of biofeedback training of these muscles introducing the intranasal placement of sEMG electrodes and a home exercise program.

Methods: A nonrandomized pilot study of patients presenting with symptoms of obstructed nasal breathing was conducted. All selected patients (n = 15) demonstrated nasal valve stenosis with a positive Cottle maneuver and clinically evident nasal valve collapse. Follow-up ranged from 9 to 12 months. Treatment included surface and intranasal EMG biofeedback-assisted specific strategies for nasal muscle education and a home exercise program of specific nasal movements.

Results: All patients exhibited variable subjective improvement. In 86.66% (n = 13), the improvement was proved objectively and there was no need for operation. In 13.33% (n = 2), an endonasal operation was recommended.

Conclusion: Relieve of obstructed nasal breathing after nasal valve stenosis and/or collapse can be achieved with biofeedback training of nasal muscles and a home exercise program as described. It helps a significant cohort of patients with symptoms of obstructed nasal breathing to avoid surgical intervention.

© 2005 Elsevier Inc. All rights reserved.

1. Introduction

Symptoms like nasal obstruction and congestion, which usually are described by patients as difficulties of nasal breathing associated with unpleasant sense of fullness or heaviness in the nose, may result in different outcomes from chronic mouth breath to a need for endonasal operation. Evaluating patients with nasoseptal deformities, chronic rhinitis, hay fever, acute or chronic sinusitis, nasal allergy, scarring from burns or trauma, Bell's palsy, stroke, senile atrophic changes of upper and lower cartilages and nasal muscles, and other disorders that can affect function of the nasal valve, a surgeon may suggest the operation of septorhinoplasty, implants, or suspension sutures to relieve obstructed nasal breathing.

Although the above mentioned etiology includes both anatomical and physiological changes, the nasal valve can be compromised either anatomically (stenosis) or physiologically (collapse), or by stenosis + collapse combined. The nasal valve, first described by Mink in 1903 and further investigated by Haight and Cole in 1983 [1,2], is the narrow passage between the lower margin of the upper lateral cartilage and the septum. It controls the inspired air currents, changing them from a column to a sheet of air and giving them both resistance and velocity, producing greater depth of respiration. In case of collapse, the role of the muscles of

* Corresponding author. Department of Otolaryngology, Assaf Harofeh Medical Center, Tel Aviv University, Tel Aviv 70300, Israel. Tel.: +972 3 553 6139; fax: +972 3 553 6137.

E-mail address: vaimed@yahoo.com (M. Vaiman).

[1] Affiliated with Sackler Faculty of Medicine.

0196-0709/$ – see front matter © 2005 Elsevier Inc. All rights reserved.
doi:10.1016/j.amjoto.2005.02.004

the nose is very important and should be evaluated very carefully. If the nasal valve is narrowed by weakness of these muscles, it is predisposed to collapse prematurely, resulting in symptoms of nasal blockage. As it was previously investigated [3], when the alar muscles were paralyzed by lidocaine block of the nerve VII, alar collapse occurred. In natural way, the same can be observed after stroke or as a result of the Bell's palsy.

Although rigid nasal valve stenosis due, for example, to scars is to be treated only surgically or with supportive devices, in cases of nasal valve collapse, the surgical intervention can be avoided if the patient will be able to strengthen his nasal muscles responsible for the shape of nasal valve. This can be achieved by nasal neuromuscular training using visual electromyographic (EMG) feedback and a home exercise program of specific facial movements.

Biofeedback EMG instruments are essentially general purpose physiological monitoring devices that are designed to provide ongoing information about a physiological function, such as muscle tension level as measured by EMG. To be useful, this information should be presented in a form that is easy for both the therapist and the patient to observe and understand. The process of monitoring and "feeding back" information to the patient is used to aid in training the patient to alter some characteristic of that activity.

Facial neuromuscular training with EMG biofeedback is well described [4-7]. At the same time, we did not find in literature any description of treatment using EMG feedback with a special emphasis to nasal muscles or to endonasal problems. The purpose of this paper is to describe the outcome of biofeedback training of nasal muscles in cases of nasal valve stenosis and/or collapse. The present study was performed to investigate the best way of using sEMG in biofeedback training of muscles involved in nasal valve function. In the present study, we tried to improve the way of biofeedback training of these muscles introducing the intranasal placement of sEMG electrodes and a home exercise program of specific nasal movements.

2. Materials and methods

2.1. Subjects

The patients were selected across a 3-month period. Follow-up ranged from 9 to 12 months. The study was approved by the Medical Center Ethics Committee (outpatient department). The subject group included 15 adults (n = 15), all Caucasians, nonsmokers, 5 women, and 10 men, ranging in age from 18 to 55 years (mean, 27.4 years). Before the study, all subjects completed a questionnaire regarding their general health and their medical history. Subjects had deviated nasal septum (9) and nasal valve collapse because of trauma (1), operation of rhinoplasty (2), and idiopathic weakness of nasal muscles (3). All selected subjects had no nasal valve stenosis due to unchangeable anatomical alterations (functional vs anatomical inclusion criteria). All subjects were examined by ENT physicians before their participation to the study.

Enrollment procedure: assessed for eligibility (n = 48)

Excluded (n = 32)

(*a*) Not meeting inclusion criteria (had anatomical stenosis with negative Cottle maneuver, usually as a result of severe septal deviation) (n = 26)

(*b*) Refused to participate (n = 6)

Enrolled and received allocated treatment (n = 16)

Lost to follow-up because of relocation (n = 1)

Analyzed (n = 15)

The Cottle maneuver is a simple test to detect any limitations in inhalation at the level of the ostium internum, and the nasal valve is to pull the cheek lateralwise during gentle inhalation through the nose [8,9] (Fig. 1). This sign is positive when inhalation noticeably improves with this maneuver. Fifteen subjects were selected from 48 patients on the criterion of their nasal anatomical changes: patients with nasal valve collapse (or stenosis + collapse) and positive Cottle maneuver were included in the study; patients with severe nasal valve stenosis and negative Cottle maneuver were excluded.

2.2. Home exercise program of specific nasal movements

The selected patients were learned to perform 3 movements of the nose:

1. To shorten the nose; instruction given to a patient: "Wrinkle your nose without knitting your brows. Do not make a complete wry face, just wrinkle your nose." This exercise helps to train the musculus procerus, the musculus levator labii superioris alaeque nasi, the musculus anomalous nasi, and to some extent, the musculus transverse nasalis.
2. To lengthen the nose; instruction given to a patient: "Move your nostrils down without knitting your brows or moving your lips and chin." This

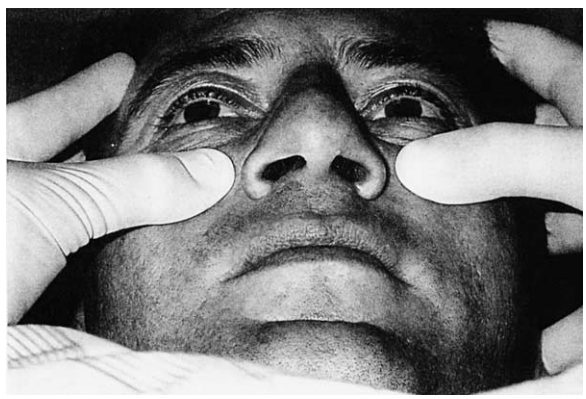

Fig. 1. The Cottle maneuver being inclusion criteria, as it was performed during the enrollment procedure.

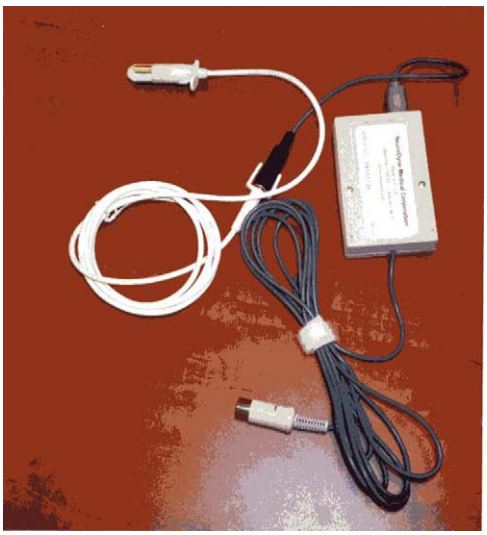

Fig. 2. An intranasal sEMG sensor.

exercise helps to train musculus nasalis (both parts), musculus depressor septi, and musculus compressor narium minor.

3. To narrow the nostrils; instruction given to a patient: "Try to dilate your nostrils without making a wry face." This exercise helps to train the alar portion of the nasalis muscle and the dilator naris anterior.

The patients were instructed to perform this program as many times a day as possible, 5 to 10 minutes each time, in front of a mirror. The movements are to be made rapidly, vigorously, and indiscriminately to breathing. These exercises help to build up muscle strength and to teach a patient how to operate his nasal muscles. The program helps patients to become nasal muscle conscious. Usually, 1 month is enough to achieve these goals. During the second month of training, the patients are trained to widen the nostrils each time he/she inhales; they continue the exercises of the first month. Starting from the third month of training, the patients perform the initial 3 exercises only once in the morning, and their main effort is to breathe with open nostrils. In ideal, this additional movement of nasal muscles becomes automatic.

2.3. Electromyographic biofeedback techniques

Seven muscles were trained in the study: musculus levator labii superioris alaeque nasi, musculus anomalous nasi, musculus nasalis, musculus dilator naris posterior, musculus depressor septi, musculus dilator naris anterior, musculus compressor narium minor. All these muscles are superficial and they are involved in nasal valve movements.

All sEMG recordings were made using standard surface sensors AE-178 (NeuroDyne Medical Corp). We used Neuromuscular Sys/3 4 channel computer-based EMG system with NeuroDyne Medical software (NeuroDyne Medical Corp, Cambridge, MA) and AE-204 Active Electrodes (NeuroDyne Medical Corp). The system uses the Active Electrode, a compact sensor assembly that includes a miniaturized instrument preamplifier. Locating the amplifier at the electrode site allows the artifacts to be cancelled and the signal boosted before being transferred down the electrode cable. Each EMG record was full-wave rectified and low-passed filtered. The computer program indicates mean, SD, minimum and maximum range of muscle activity during each trial, and its duration. Muscle activity (EMG) is quantified in microvolts.

For intranasal EMG biofeedback, we invented intranasal EMG sensor. This sensor (Fig. 2) is a tube that has is oblong designed to fill the nasal vestibule up to the nasal

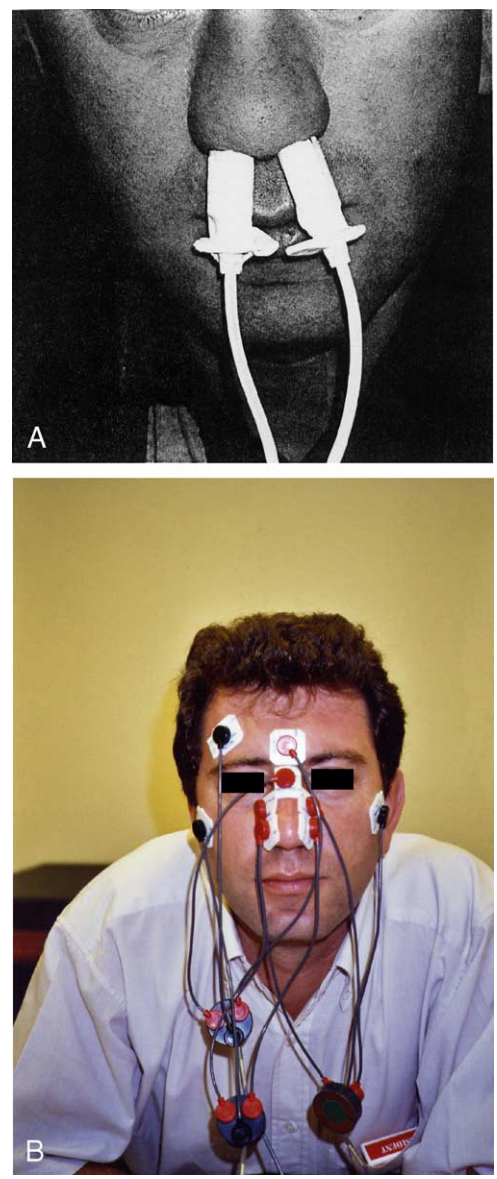

Fig. 3. (A) A subject with 2 intranasal EMG sensors, as used during biofeedback training. Each sensor occupies one channel of the EMG device. (B) A subject with 3 sEMG sensors, as used during biofeedback training. Each sensor occupies one channel of the EMG device. The fourth channel can be used for the intranasal sensor.

Name: E
Protocol: Single EMG - Increase
Session Date: 29 Nov 2002
Session Time: 2:28 PM

Measure: EMG 1
Sensor: EMG 1 Site: m. nasalis

Trial	Mean	StdDev	Minimum	Maximum	Goal	Success	Thresh.	Criterion
Base	4.019	0.781	2.989	7.583				
1	11.343	5.036	4.264	29.756	Up	100%	3.899	97%
2	16.992	5.594	7.185	31.548	Up	100%	7.583	189%
3	25.037	7.232	3.811	54.361	Up	96%	12.210	304%
4	32.003	8.168	3.627	55.848	Up	95%	18.630	464%
5	41.697	9.455	7.315	78.251	Up	97%	25.291	629%
6	45.157	11.893	7.348	85.231	Up	86%	33.419	832%
7	42.933	10.906	11.164	73.807	Up	83%	33.419	832%

Fig. 4. An example of a biofeedback training report, one channel, goal—tension increase. One pair of sEMG electrodes was placed above nasal muscles. Baseline during normal breathing was established as mean 4.019 μV. During trials 1 to 6, the patient was able to increase the tension of her nasal muscles up to 85.231 μV maximum with mean 45.157 $\pm$ 11.893 μV. Then, during trial 7, the patient got tired and the session was discontinued.

valve. It has 2 active electrodes and 1 ground electrode, and a wide breathing hole in a center. The diameter goes in small (10 mm), medium (12 mm), and large (14 mm) sizes. To achieve better electrode contact and to prevent mucosa from scratching, the intranasal sensor should be used with an electrode gel suitable for contact with mucosa (highly conductive multipurpose electrolyte "Signa-gel" electrode gel, Parker Labs, Inc, NJ, or similar). Each intranasal EMG record also was full-wave rectified and low-passed filtered. This rectified and filtered record looks like a single line, similar to an electrocardiogram trace, easy to observe and understand.

Electromyographic biofeedback consisted of 3 interrelated activities: (1) initial assessment of the amplitude (range, voltage) and graphic pattern of activity of above-mentioned nasal muscles, (2) providing patients with feedback information about their physiological activity to aid them in learning to increase the amplitude and improve the graphic pattern, and (3) recording the process outcome of training using the physiological measures as objective data.

2.4. Procedures

After electrode placement, each participant performed 3 tasks:

1. Normal nasal breathing; instruction given to a volunteer: "Breathe normally"—helps to establish the baseline.
2. Forced nasal inspiration; instruction given to a volunteer: "Breathe as deep as you can but in normal pace"—helps to demonstrate changes in amplitude of muscle activity (curve tracing) during breathing to the patient.
3. Breathing with widening of the nostrils; instruction given to a volunteer: "Try to move your nostrils to make them wider when you inhale, you should feel an increase of airflow and easy breathing"—helps to train the patient to breathe with open nasal valve.

After that, the graphic patterns were analyzed with patient, correction given, explanations provided. For the main exercise, the external surface electrodes were removed except a pair above the nostrils, and subjects performed the training with this surface pair (1 channel) and with an intranasal electrodes gently inserted (1 or 2 channels) (Fig. 3A and B). Sometimes, when a patient was confused with 2-channel recording, we used just 1 channel EMG record. When sensors were attached and setup period was completed, a site location was indicated and baseline values (current level of activity of nasal muscles) were established by the computer program. We used 3 site locations: left nostril (intranasal), right nostril (intranasal), musculus nasalis and musculus dilator naris anterior (surface). The goal setting was for increasing nasal muscle tension levels, that is, to increase an amplitude of the recorded curve. The amplitude represented strength of the muscles. The EMG protocol had goal and baseline display options, and lines representing the trial goal and the baseline reference appeared on the screen of computer when the training trials began. If the patient failed to increase the tension of his nasal muscles at least 50% of the time during a trial, the goal was adjusted to be easier at the start of the next trial. For success between 50% and 90%, the goal was left the same. For success above 90%, the goal was adjusted to be made more challenging for the patient (new goal = [old goal + mean value for trial]/2). Each trial lasted for 3 minutes.

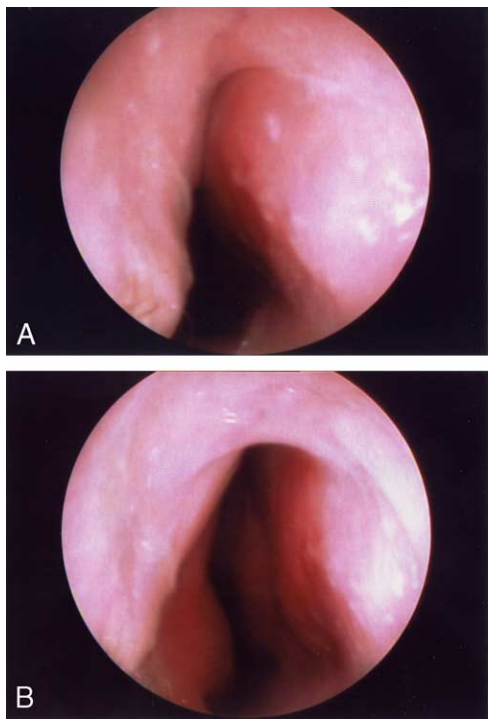

Fig. 5. Endonasal pictures of nasal valve before (A) and after (B) treatment. Follow-up after 9 months.

Single site protocol 1: trial 1, left nostril; trial 2, surface location; trial 3, right nostril; trial 4, surface location.

Single site protocol 2: single location is taken for several trials until the patient is tired (usually 4–8 trials) (Fig. 4).

Two-site protocol: trial 1, two nostrils; trial 2, left nostril + surface location; trial 3, right nostril + surface location; trial 4, 2 nostrils.

The curve tracing feature was used for EMG respiration training. It was used to train the patient to form a graph having a particular shape of a respiration with well controlled nasal muscles and nasal valve wide open. For neuromuscular education, this was useful in giving the patient control over fine nasal movements, rather than just a coarse contraction. It was used to teach the patient to adjust nasal muscles contraction with normal breathing pace.

Patients performed biofeedback training 3 times a week during first 6 weeks. Afterward, they had one training a week for the next 6 weeks. During further follow-up visits, the EMG records of nasal muscle activity was obtained every month.

3. Results

Subjective self-assessment data for nasal airflow were collected. Objective data were provided by

- pretreatment and posttreatment observation;
- photographic analysis (nasal endoscopy) during the treatment and follow-ups (Fig. 5A and B);
- anterior rhinomanometric evaluation (Rhinomanometer NR6, GM Instruments, Scotland, UK) at a fixed pressure of 150 Pa, with technical arrangements and data calculations made according to the recommendations of the International Standardization Committee for Rhinomanometer [9];
- acoustic rhinometry evaluation (GMI A1 Acoustic Rhinometer, GM Instruments) with established technical arrangements [10];

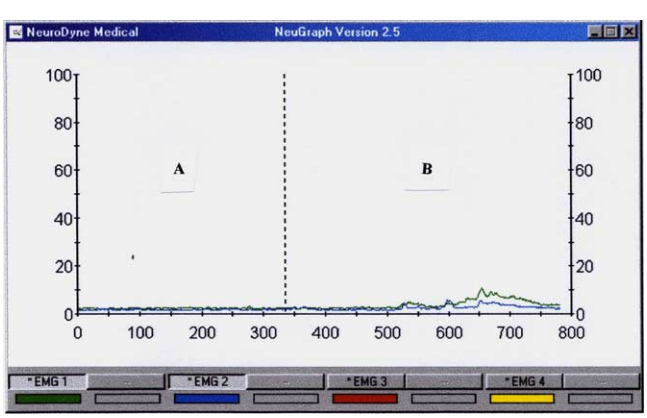

Fig. 6. An EMG 2-channel (surface + intranasal) record of breathing of a patient with symptoms of obstructed nasal breathing and clinically evident nasal valve collapse. A indicates normal breathing; B, deep breathing. The patient is almost incapable to operate his nasal muscles. Intranasal sensor record is slightly above the record from the surface electrode.

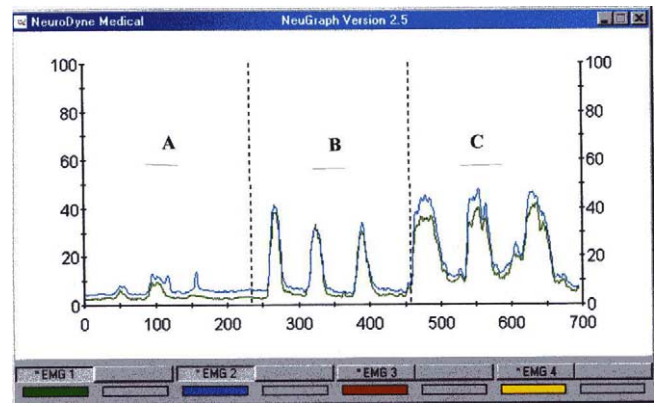

Fig. 7. The same patient after complete course of EMG biofeedback training and home exercise program, follow-up visit after 10 months. A indicates quiet normal breathing; B, normal "nasal-valve-open" breathing; C, deep breathing. Intranasal sensor record is slightly above the record from the surface electrode.

- surface and intranasal EMG records before or during the treatment and follow-ups (NS/3 EMG, NeuroDyne) (Figs. 6 and 7).

Multivariate analyses of variance for repeated measures were used to analyze the results before and after training (mean interval, 41.3 weeks). For acoustic rhinometry and rhinomanometry (Table 1), data were also analyzed using the *t* test. Values of $P < .01$ were considered significant. After treatment, 13 patients (n = 13, 86.66%) had increased airflow, that is, bilateral inspiratory nasal flow increased ($P < .001$). All patients reported subjective improvement to some extent. Nasal inspiratory resistance significantly decreased ($P < .0001$) in 12 patients (n = 12, 80%).

In acoustic rhinometry, first minimal cross-sectional area (1.Amin) represents cross-section of nasal valve. Before treatment, this area varied among subjects from 0.271 to 0.638 cm^2 (mean, 0.493 cm^2; SD, 0.17). First minimal cross-sectional area in the nasal cavities (actually, nasal valve area) significantly increased in 11 patients (n = 11, 73.47%) with mean value of 0.842 cm^2 and SD of 0.263. Although the other 4 patients also presented some increase of cross-sectional area of nasal valve, this improvement was not statistically significant.

Electromyography-recorded amplitude of muscle tension of the nasal muscles (in microvolts) significantly increased in all 15 patients (100%) ($P < .001$), that is, strength of the nasal muscles was increased (Table 2). The amplitude data taken from the intranasal sensor were, as a rule, higher than

Table 1
Statistical analysis of rhinomanometric data before, during, and after treatment

	Media	SD	*P*	t
Before treatment	0.62	0.153		
After 1 month of treatment	0.25	0.092	<.0001	8.4
After 3 months of treatment	0.18	0.078	<.0001	14.1
Follow-up after 6 months	0.18	0.120	<.0001	14.1
Follow-up after 9 months	0.19	0.095	<.0001	13.55

Table 2
Real mean of electric activity of nasal muscles (in microvolts) recorded during normal breathing from the skin location (musculus transverse nasalis) and from intranasal sensor (musculus dilator naris anterior and musculus alar nasalis)

	Skin location	Intranasal location
Before treatment	4.35 ± 2.86	6.03 ± 1.3
After 1 month of treatment	7.89 ± 2.95 ($P < .01$)	12.6 ± 2.65 ($P < .005$)
After 3 months of treatment	10.06 ± 3.64 ($P < .001$)	15.02 ± 3.72 ($P < .001$)
Follow-up after 6 months	23.01 ± 5.54 ($P < .001$)	26.81 ± 6.89 ($P < .001$)
Follow-up after 9 months	19.42 ± 4.86 ($P < .001$)	23.22 ± 5.88 ($P < .001$)

Real mean = Computer calculated mean of a trial (raw) − 2.82 μV (Mean electric tension of the relaxed nasal muscles) or − 2.15 μV (Mean electric tension of the relaxed nasal muscles recorded from intranasal sensor).

those taken from skin location, but the tendency for increase of the voltage was practically the same.

Two patients (n = 2, 13.33%) did not show significant objective improvement of the nasal functions and were recommended for endonasal surgery.

4. Discussion

In case of facial muscle problems, EMG feedback is used mostly in the treatment of facial nerve paralysis [11-13] and is not recommended to patients with facial paralysis after stroke [14]. We first used EMG biofeedback-assisted specific protocols for nasal muscle training and a home exercise program of specific nasal movements to treat endonasal disorders. We found this technique very useful in treatment of the patients with symptoms of obstructed nasal breathing who wish to avoid surgical operation or supportive devices.

It is clear that the procerus, the anomalous nasi, and the transverse part of the musculus nasalis do not affect the function of the nasal valve because they are located above the nasal bone part and the upper lateral cartilage. At the same time, the dilator naris anterior and the alar part of the musculus nasalis are very active in nasal valve regulation in case if they are developed well. The direct influence of the depressor septi and the levator labii superioris alaeque nasi on the nasal valve is not yet clear. In any case, we advise to train all muscles of the nose.

The intranasal sensor was used to introduce different variations of the training, aiming to involve as many nasal muscles as possible, and at the same time, to avoid involvement of other groups of facial muscles. Our patients reported that using of intranasal sensor helped them to perform training because they felt a bearing inside the nose to support alae of nostrils. The intranasal sensor is a simple and nonexpensive device, well compatible with existing IA-250 EMG amplifier (NeuroDyne Medical Corp) and sEMG devices. We found it well fit for both screening procedures and EMG biofeedback treatment. When the 2 intranasal sensors are inserted, they can serve for quick objectivization of left-right nostril differences in cases of stroke, Bell's palsy, septal deformity, and similar. In biofeedback training, this objectivization should be used during initial explanation of the problem to a patient.

In case the patient continues using the exercise program routinely, the results, as we observed, remain good for a long time. Even when the follow-up period is over, we usually recommend patients to visit the biofeedback training room once a month or once every 2 months to track changes in the muscle strength and coordination.

5. Conclusion

We conclude that significant relieve of nasal valve stenosis and collapse can be achieved with biofeedback training of nasal muscles and a home exercise program as described. It helps a significant cohort of patients with symptoms of obstructed nasal breathing to avoid surgical intervention. The newly introduced intranasal sensor helps EMG biofeedback training to be more effective. The method is simple, noninvasive, and has good results.

References

[1] Mink PJ. The nose as a means of breathing. Presse Otolaryngol Belg 1903;21:481-96.
[2] Haight JS, Cole P. The site and function of the nasal valve. Laryngoscope 1983;93:49-55.
[3] Brach JS, VanSwearingen JM, Lenert J, et al. Facial neuromuscular retraining for oral synkinesis. Plast Reconstr Surg 1997;1922-31.
[4] Shiau J, Segal B, Danys I, et al. Long-term effects of neuromuscular rehabilitation of chronic facial paralysis. J Otolaryngol 1995;24(4):217-20.
[5] Segal B, Hunter T, Danys I, et al. Minimizing synkinesis during rehabilitation of the paralyzed face: preliminary assessment of a new small-movement therapy. J Otolaryngol 1995;24(3):149-53.
[6] Duckett S, Kramer T. Managing myoclonus secondary to anoxic encephalopathy through EMG biofeedback. Brain Inj 1994;8(2):185-8.
[7] Cottle MH, Loring RM, Gaynon IE. Correlating nasal examination with respiratory tract function. AMA Meeting, San Francisco; 1958. p. 23-7.
[8] Nolst Trenite GJ. Rhinoplasty. The Hague, the Netherlands: Kugler Publications; 1998.
[9] Clement PAR. Committee report on standardization of rhinomanometry. Rhinology 1984;22:151-5.
[10] Hilberg O, Jackson AC, Swift DL, et al. Acoustic rhinometry: evaluation of nasal cavity geometry by acoustic reflection. J Appl Physiol 1989;66:295-303.
[11] Diels HJ. Treatment of facial paralysis using electromyographzic feedback—a case study. Eur Arch Otolaryngol Suppl 1994 S129-32.
[12] Segal B, Zompa I, Danys I, et al. Symmetry and synkinesis during rehabilitation of unilateral facial paralysis. J Otolaryngol 1995;24(3):143-8.
[13] Gallegos X, Medina R, Espinoza E, et al. Electromyographic feedback in the treatment of bilateral facial paralysis: a case study. J Behav Med 1992;15:533-9.
[14] Svensson BH, Christiansen LS, Jepsen E. Treatment of facial nerve paralysis with electromyography biofeedback and taping of cheek. A controlled clinical trial. Ugeskr Laeger 1992;154:3593-6.